

Ismail AISSAOUI

Artificial Intelligence & Data Science Engineer

Email: ismail.aissaoui.pro@gmail.com

Phone: +213 660 70 77 96

Location: Algeria

GitHub: github.com/i-aissaoui

To Professor Carlos Crispim-Junior
LIRIS - Université Lumière Lyon 2
Bron, France
May 7, 2026

Subject: Application for PhD Position: Emotion Understanding in Videos (AIMAINT Project)

Dear Professor Crispim-Junior,

I am writing to express my strong enthusiasm for the PhD position in Artificial Intelligence and Computer Vision at LIRIS, focused on emotion understanding in videos. As an AI Engineer from ESI-SBA with a deep background in real-time facial recognition and sequential modeling, I am particularly motivated by the AIMAINT project's goal of leveraging Computer Vision to support ASD diagnosis and behavioral analysis.

My technical profile aligns precisely with the challenges of emotion recognition in unconstrained medical settings:

- **Facial Recognition & FER:** Through my project **Recognizini**, I developed a real-time facial recognition system that incorporates active feedback loops and incremental embedding updates. This experience gave me a deep understanding of facial feature extraction and the robustness required for varying illumination and head poses.
- **Video Representation & Sequential Modeling:** My Master's thesis at **Sonatrach** focused on the use of **Mamba-SSM** and **xLSTM** for modeling long-term temporal dependencies in industrial telemetry. I am eager to apply these state-of-the-art sequential architectures to the problem of **emotion synchrony**, where understanding the temporal flow of micro-expressions is critical for accurate behavioral coding.
- **Domain Generalization:** I have extensive experience in building models that generalize across heterogeneous datasets. My implementation of **Vision Transformers (ViT)** and my work on the N-CMAPSS fleet dataset have prepared me to tackle the challenge of adapting foundation models to the subtle, low-intensity emotional expressions found in clinical environments.

Beyond technical expertise, I am deeply committed to the ethical application of AI for social good. My recent application to the French Social Ministries (DNUM) reflects my desire to use data science for public health and protection. Joining the LIRIS laboratory would allow me to combine my Computer Vision skills with meaningful healthcare impact.

I am proficient in Python, PyTorch, and OpenCV, and I am accustomed to working in interdisciplinary contexts. I would welcome the opportunity to discuss how my research background can contribute to the AIMAINT project's success.

Thank you for your time and consideration.

Sincerely,

Ismail AISSAOUI